ExploreMate – Project Documentation

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1. Introduction

ExploreMate is an AI-powered travel companion application designed to enhance the overall travel experience. Unlike traditional travel applications that focus mainly on booking and navigation, ExploreMate provides personalized, interactive, and intelligent features that assist users during both trip planning and real-time exploration.

The application is especially useful for solo travelers and introverts, allowing them to explore places independently without relying on guides or external assistance.

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2. Objectives

- To provide personalized travel recommendations
- To help users discover hidden and unexplored places
- To enhance user engagement through gamification
- To assist users with AI-based guidance and audio storytelling
- To simplify travel planning and communication

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3. Feature Classification

General Features

These are common features that support the application:

- Smart Search
- AI Assist
- Translator
- Trip Scheduler

Key Features (Core USP)

These features differentiate ExploreMate from other applications:

- Hidden Gem Finder
- Mood-Based Food Explorer
- Gamified City Explorer

- AI Audio Tour Guide

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4. General Features

4.1 Smart Search

Allows users to quickly find places, food, and hidden gems.

Flow:

- User enters keywords (e.g., “beach”, “pizza”, “vizag”)
- Backend searches across categories
- Results and suggestions are displayed

4.2 AI Assist

Acts as a virtual guide to answer user queries.

Functionality:

- Basic Q/A system
- Provides suggestions and recommendations

4.3 Translator

Helps users overcome language barriers.

Features:

- Text-based translation
- Camera-based translation (OCR + Translate)

Flow:
Camera → Text Extraction → Translation → Display

4.4 Trip Scheduler

Helps users plan their trips efficiently.

Flow:

- Input: destination, duration, preferences
- Output: day-wise itinerary

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5. Key Features

5.1 Hidden Gem Finder

Helps users discover underrated and locally popular places.

Data Stored:

- Place name
- Location
- Description
- Image

Output:
List view → Detail view

5.2 Mood-Based Food Explorer

Provides food recommendations based on user mood and weather.

Flow:

- User selects mood
- Backend filters results
- Displays relevant food places

5.3 Gamified City Explorer

Enhances engagement by turning travel into a game.

Components:

- XP System
- Levels (Beginner → Pro Traveler)
- Badges (e.g., Hidden Hunter, Foodie)
- Missions (tasks with rewards)
- Leaderboard

5.4 AI Audio Tour Guide

Provides audio-based guidance for places.

Flow:

- User selects or enters place
- AI generates description
- Text converted to audio
- Audio is played

Additional Feature:

- Q/A box for user queries

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6. AI Integration

The application uses AI to dynamically generate content.

- AI generates place descriptions
- Text-to-Speech converts content into audio
- Supports dynamic and scalable content generation

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7. Authentication System

Registration:

- Name
- Email
- Password
- Phone Number
- OTP Verification (Firebase)

Login:

- Email/Username + Password

Session:

- Auto-login enabled unless user logs out

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8. Technology Stack

Frontend:

- Flutter

Backend:

- Node.js
- Express.js

Database:

- PostgreSQL

Authentication & Notifications:

- Firebase

APIs Used:

- Google Maps API
- Weather API
- Translator API
- Text-to-Speech API
- OpenAI (for AI features)

Deployment:

- Backend: Render
- App: APK / Mobile

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9. System Architecture

Flutter App
↓
Firebase Authentication (OTP)
↓
Node.js Backend
↓
PostgreSQL Database
↓
External APIs (Maps, AI, etc.)

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10. Team Structure

1. Backend Controller

- Architecture design

- API development
- Database design

2. Backend Developer (Support)

- Data handling
- API testing
- Error handling

3. Flutter Lead (UI)

- UI development
- Screen flow
- API integration

4. Integration & Firebase

- OTP setup
- API integration
- Error management

5. Testing & Presentation

- Testing
- Demo preparation
- PPT

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11. Conclusion

ExploreMate is a comprehensive travel application that combines AI, personalization, and gamification to deliver an enhanced and immersive travel experience. By integrating intelligent features with user-friendly design, the application aims to redefine how users explore and experience new places.